

Supporting Information:

Supporting Information for MARCIAH: A Q6 Workflow for Protein–Ligand Linear Interaction Energy Calculations

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1 Minimal Command-Line Test

The executable MARCIAH distribution provides a minimal command-line test. The test is launched through `scripts/run_cli_example_test.sh`, uses the bundled ligand and complex production logs, runs `analyze_LIE_noqgui.py`, and writes a CSV file with the resulting LIE estimate under `results/`.

2 Software Environment

The Python-side dependencies used by the analysis scripts are defined in the root-level `environment.yml` file of the MARCIAH repository. Installation is automated through `scripts/setup_conda_env.sh`, which creates the `marciah-q6-lie` Conda environment.

3 Required File Classes

The workflow requires ligand and complex inputs that are compatible with Q6. At minimum, one calculation uses ligand and complex topology files, ligand library and parameter definitions, FEP files, equilibration restart files or the inputs needed to generate them, production input files, and production log files for the final LIE analysis step.

4 Directory Schema

The executable repository is organized around `ligands/`, `complex/`, root-level analysis scripts, and `scripts/` for installation and CLI testing. The root `README.md` documents the execution sequence used by the bundled example.